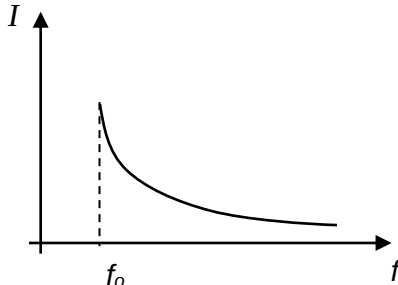
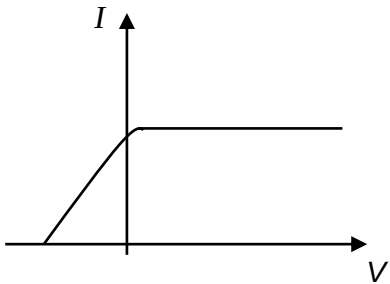
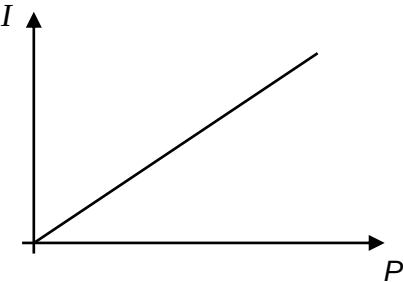


Qns	Answer	Marks
4(a)(i)	A beam of white light consists of photons of all wavelengths (λ) in the visible spectrum. The photons interact with electrons in the gas atoms OR The photons will be absorbed by the electrons (Option 1) Photon energy causes electron to move to higher energy level (or to be excited) The photon energy (of certain wavelengths ($E = hf = hc/\lambda$)) is equal to the energy difference between the energy levels of the gas atoms When electrons de-excite, photons are re-emitted in all directions (Option 2) This results in <u>fewer photons of these wavelengths/photons with much less intensity in the direction of the incident light/original direction of travel</u>. Hence, there will be dark lines corresponding to these wavelengths in the diffraction pattern.	B1 B1 B1 B1
4(a)(ii)	Transitions -3.40eV to -0.85eV and -3.40eV to -1.51eV	B1
4(b)(i)	f increases, so energy carried by each photon ($= hf$) increases. P is constant, so the total incident energy per unit time is constant. Hence, the number of photons per unit time must decrease. The number of photoelectrons must also decrease.  quantum yield = $\frac{\text{no. of photoelectrons emitted}}{\text{no. of incident photon}}$ is a fixed value for a given wavelength and type of metal. Correct shape – B1 Indication of f_0 – B1	
4(b)(ii)	Negative potential difference slows down the electrons, so fewer electrons can reach the collector, photocurrent decreases, to 0 if electrons with max KE cannot reach collector. Positive potential difference accelerates the electrons towards the collector, so all electrons reach the collector, saturation current is reached. 	B1

4(b)(iii)	With energy per photon ($= hf$) constant, the number of incident photons increases linearly with power. Hence, the number of photoelectrons also increases linearly with power. 	B1
4(c)(i)	no. of electrons $= 6^{10} = 6.05 \times 10^7$	B1
4(c)(ii)	no. of electrons per unit time from 10th dynode $= \frac{9.2 \times 10^{-6}}{1.6 \times 10^{-19}} = 5.75 \times 10^{13}$	M1
	no. of electrons per unit time from cathode $= \frac{5.75 \times 10^{13}}{6.05 \times 10^7} = 9.50 \times 10^5$	A1
4(c)(iii)	no. of incident photons per unit time $= 9.50 \times 10^5 \times 3 = 2.85 \times 10^6$	M1
	power of incident light $= n \frac{hc}{\lambda}$ $= 2.85 \times 10^6 \times \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{361 \times 10^{-9}} = 1.57 \times 10^{-12} \text{ W}$	A1
4(c)(iv)	kinetic energy of an electron $= qV = 1.6 \times 10^{-19} \times 50 = 8.0 \times 10^{-18} \text{ J}$	M1
	$E = \frac{p^2}{2m} = 8.0 \times 10^{-18} \text{ J}$	
	$\Rightarrow p = \sqrt{2mE} = \sqrt{2 \times 9.11 \times 10^{-31} \times 8.0 \times 10^{-18}} = 3.82 \times 10^{-24} \text{ N s}$ $\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{3.82 \times 10^{-24}} = 1.74 \times 10^{-10} \text{ m}$	M1 A1

Qns	Answer	Marks
Exi	Binding energy of Mo 0.5×0.00 760.55 MeV	C1